preservation

of

fodder

Preservation of **dry fodder**	Preserved **on ground**
	Exposed to air and sun
	Protected from rain

Preservation of **green fodder**	Stored under **anaerobic** condition on **acidification**

Preservation of
dry fodder

Preservation of
green fodder

SILAGE …

means preserved green fodder

which **green fodder** can be preserved ?

preferably **single cut** cereal crops (Non-Legume) or/& **legume** crops

MAKING OF SILAGE

SILAGE …

is the feedstuff resulting from preserved green fodder in **anaerobic** conditions under **acidification.**

Anaerobic conditions are created by fine chopping & compaction

ADVANTAGES OF MAKING SILAGE

- Can feed animals with **uniformity** throughout the year, which helps animals for better production & reproduction
- Can preserve **single cut** crops like maize
- Can maintain same **nutritive** values
- Can make **mixed silage** also adding maize and lucerne @ 3:1
- Can store for **2** years if unopened

REQUIREMENTS

1. Cereal crop (maize/sorghum)
2. Silo pit
3. Polyethylene sheet (200 GSM)
4. Chaff cutter
5. Silage additives

1. CEREAL crop….

preferred crop

is

whole maize crop

(African tall)

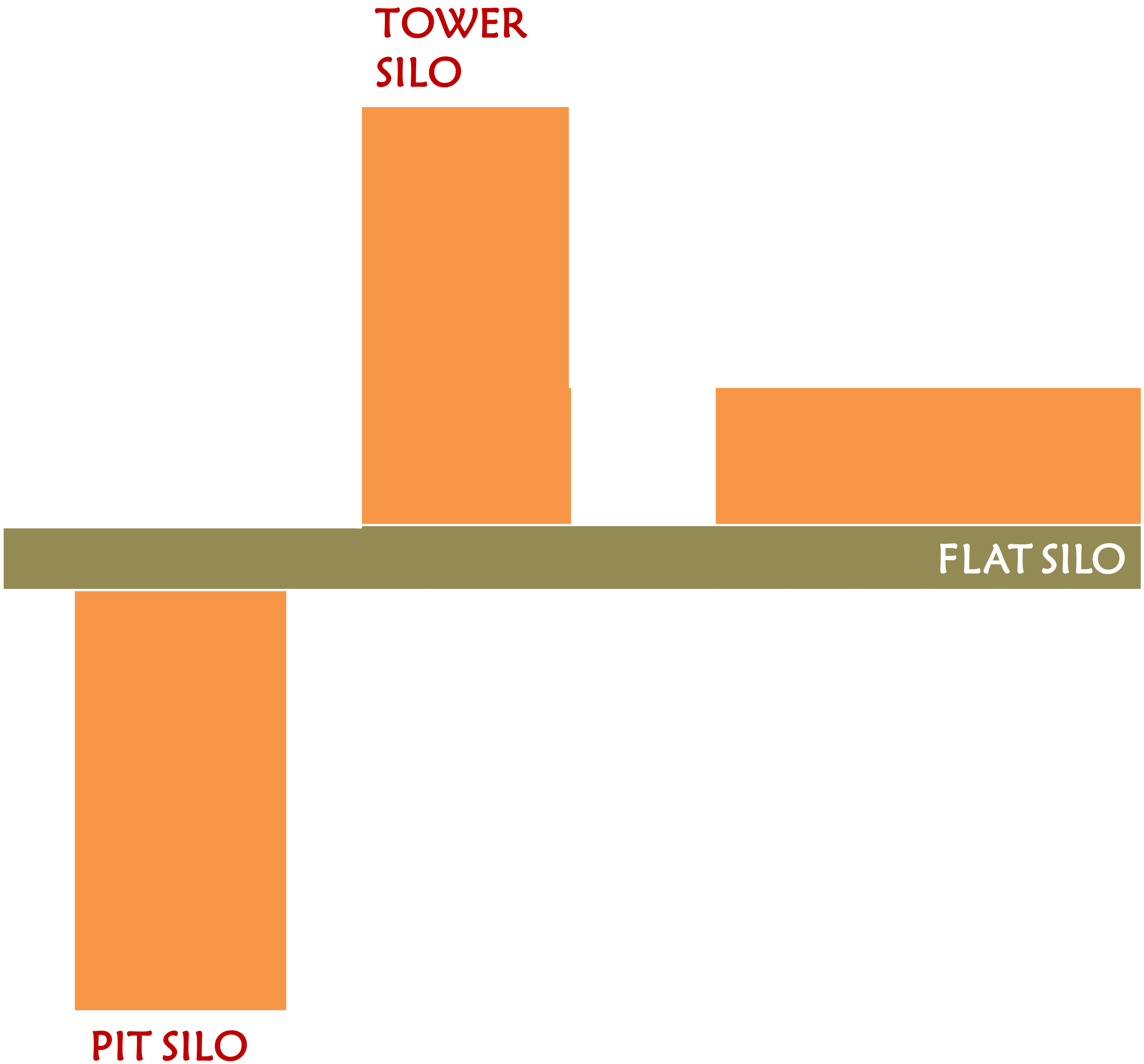
REASONS FOR SELECTING MAIZE

It contains

- high level of **fermentable sugars**
- Low level of protein
- Low buffering capacity
- Dry matter not less than 1/3 rd

SILO PITS... are two types

1. Constructed **below** the ground.....**pit** silos
2. Constructed **above** the ground... **bunker** silos
   a. **flat** silos
   b. **tower** silos


TOWER SILO
FLAT SILO
PIT SILO

PIT SILOS

advantage of pit silos is their low capital cost

well suited in Indian conditions

PIT SILO .. all the four sides & floor should be plastered with cement

PIT SILO should be 2 ft above the ground

BUNKER SILOS

Constructed above the ground ….Expensive construction cost

a. Flat silos

b.Tower silos

SIZE OF SILO PIT …depends on

- daily silage usage
  (based on the removal of a minimum of 4 inches per day)
- the total silage needed for 28-30 days

- **Depth** should not exceed **8 feet (max. 10 ft)**
- **Width** & **length** is variable

REQUIREMENT OF SILAGE FOR ONE EWE…

- Ewe needs NL-Roughage @ 2% of b.wt
- Silage contains 35% dry matter…
- A 35 kg ewe requires 700 grams of non-legume roughage@2% of b.wt
- Each ewe needs 2 kg silage per day

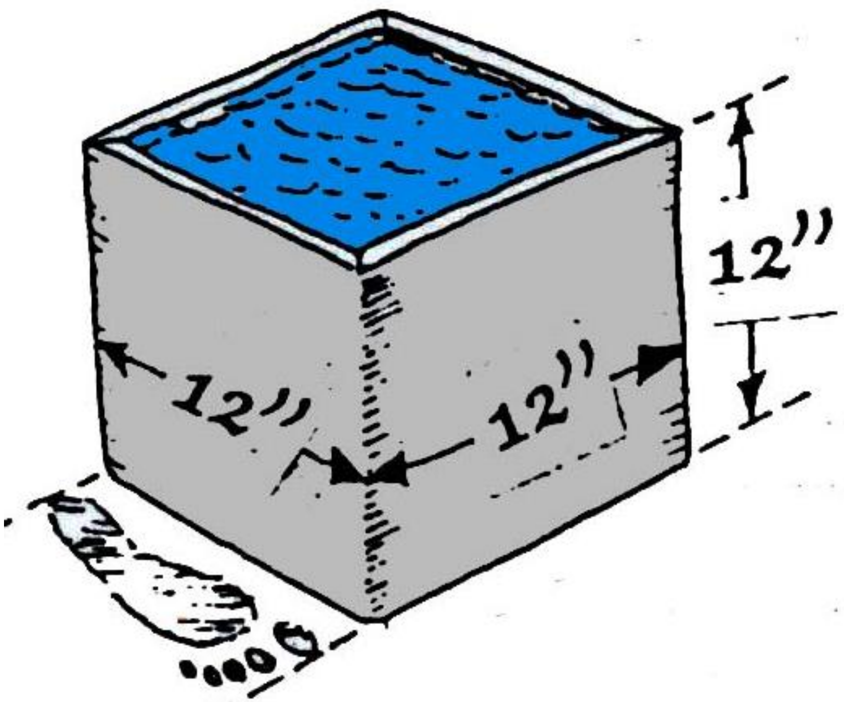
  (2 kg silage provides 700 grams of dry matter @35%)

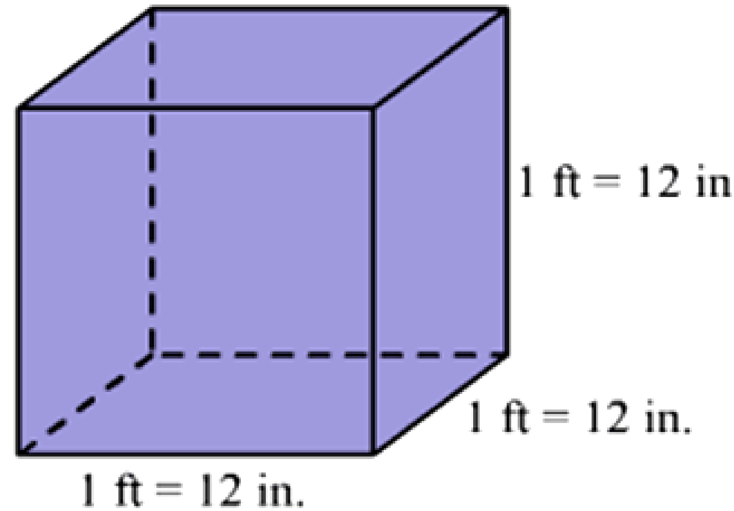
REQUIREMENT OF SILAGE FOR 20+1 UNIT

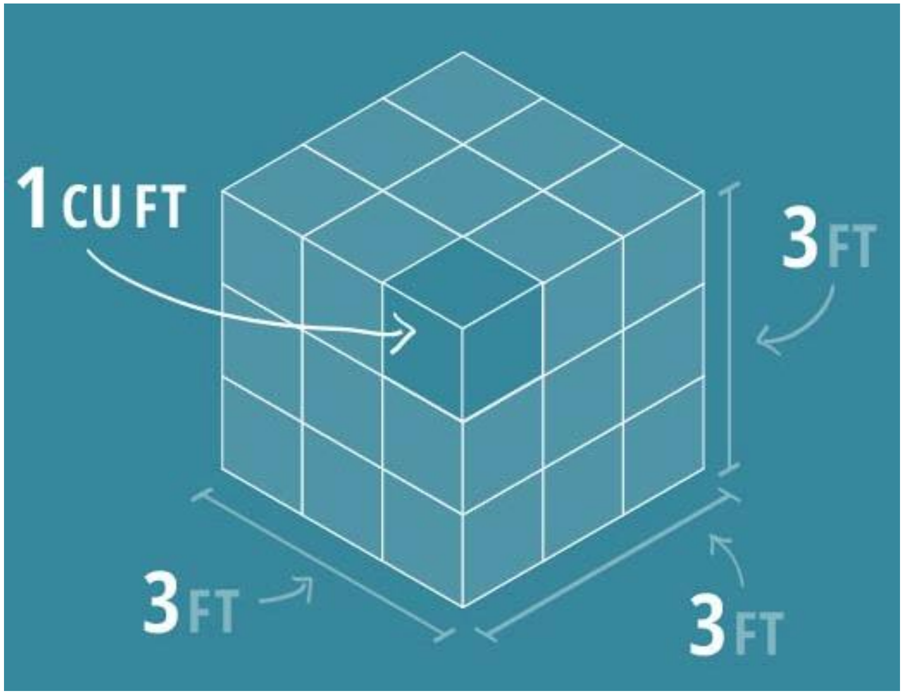
Ewes @ 2.0 kgs/day ….. (2.0x20) 40 kgs
Ram @ 2.5 kgs/day ….. (2.5x1) 2.5 kgs
Lambs @ 1.0 kg/day ….. (1x18) 18 kgs

- requirement/day 60.5 kgs
- requirement/30 days 1815 kgs (1.8 MT)
- requirement/**90** days **5445 kgs** (5.5 MT)

each cubic feet (cft) of silo pit holds 12 kgs of silage







1	2	3	4	5	6	7	8	9
2								
3								
4								
5								
6								
7								

3x3x3 = 27 cft
@ 12 kgs….holds 324 kgs

9 Length X 7 Width x 8 Depth
= 504 cft ..
holds 6048 (6 MT) of silage
@12 kg /cft

as per **monthly** requirement pit should be **partitioned** in to **equal** compartments pit should be partitioned as per the monthly requirement

3.POLYETHYLENE SHEET …

of silage grade or 200 GSM

4.CHAFF CUTTER

a. 3 blades
b. 5 HP
c. 3-4 MT chopping capacity /hour
d. movable

5.SILAGE ADDITIVES

STIMULANTS	INHIBITORS/ PRESERVATIVE	NUTRIENTS
LAB.. (Lactic acid bacillus)	salt	Urea (optional)
Molasses		DCP
		

Kem LAC® brand HD
Water Soluble Silage Inoculant
A dry, water-soluble concentrated source of microorganisms which are beneficial to silage fermentation.

INGREDIENTS: Dextrose,
Lactobacillus plantarum,
Lactobacillus acidophilus,
Lactobacillus bulgaricus,
Sodium Silico Aluminate,
Sodium Thiosulfate,
and FD&C Blue No. 1.

if molasses is not available…?

if LAB is not available ..?

ENSILING ....

the process of making silage

CHECK LIST

- be sure of **age** of forage plant
- assess **dry matter** of forage
- select **non-rainy** days
- be sure of continuous **power** supply to run chaff cutter

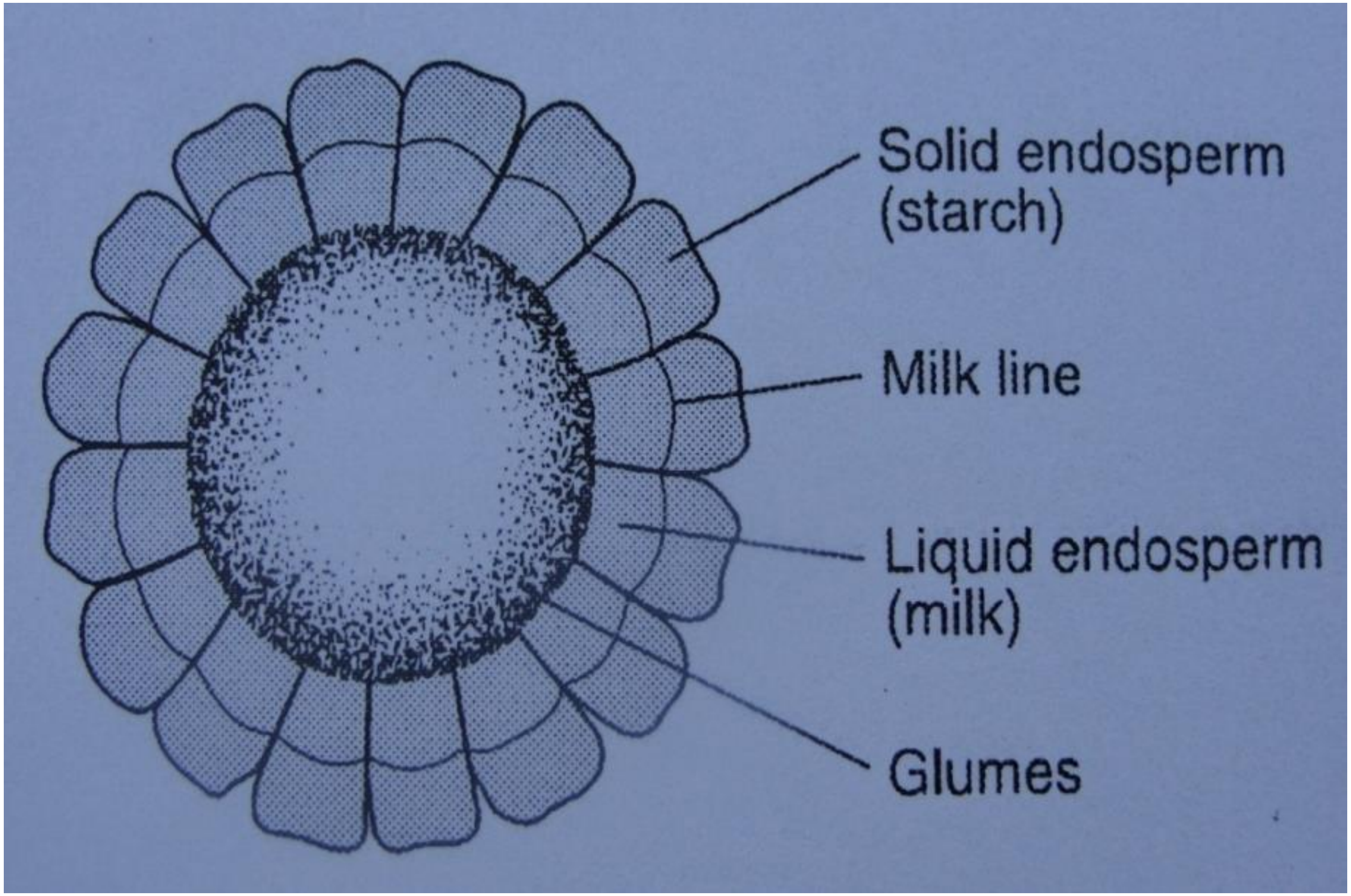
CHECK LIST

- be sure of **yield** of forage
- be sure of pit capacity
- make **additive mixture** as per requirement
- Chopping size ..should be 1 inch only
- try to seal pit during day light only

STAGE & AGE OF MAIZE CROP

(in second milk line stage or at 75-85 days age)

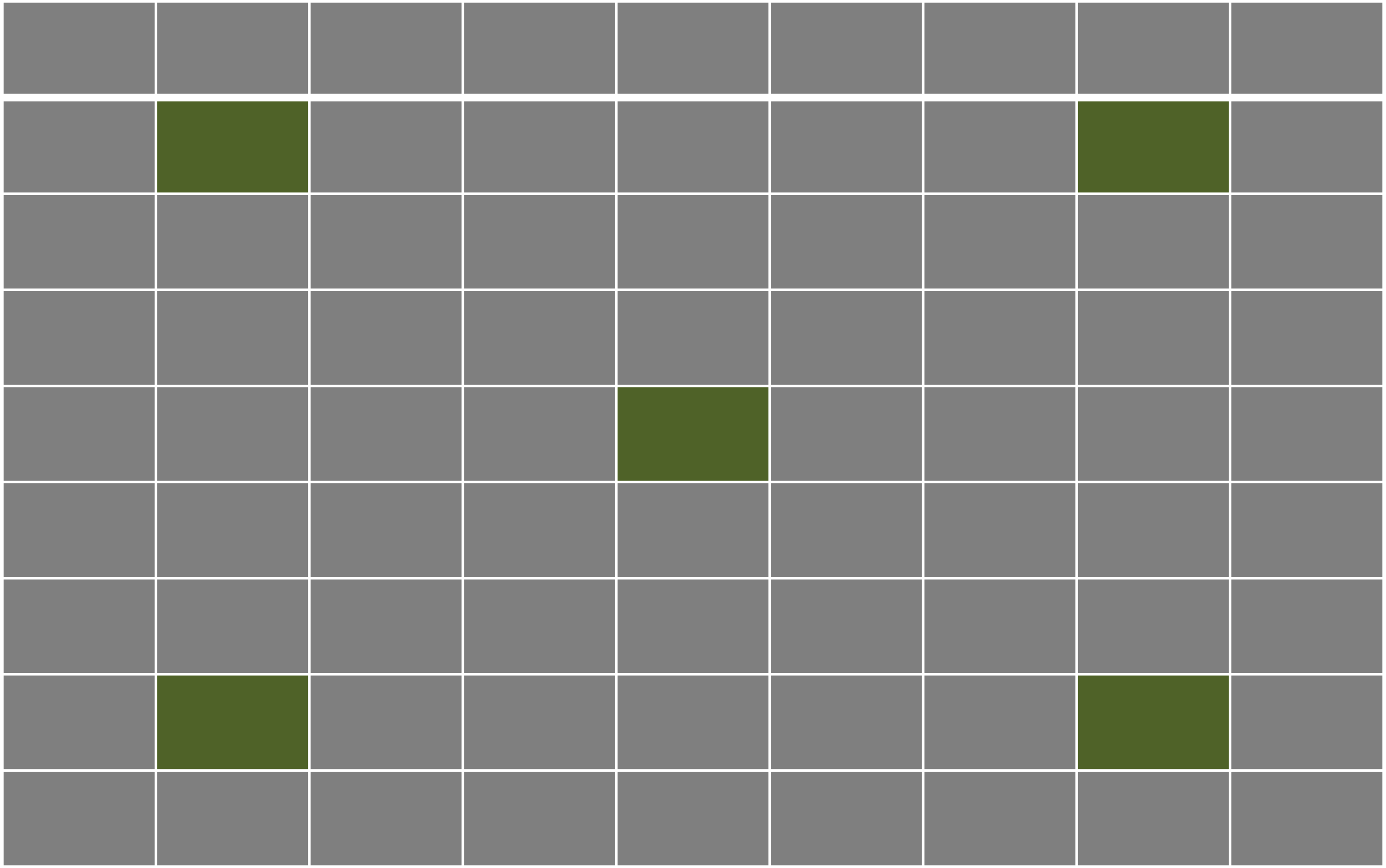
CROSS SECTION OF A CORN COB



As corn silage matures, the milk line moves down the kernel and plant composition and energy values will change.

ASSESSING YIELD OF FORAGE

- Select 5 fields in crop field as shown in picture with equal gap
- Each field should be of 11x11 ft size (121 sft)



- Take average no of plants in **each field**
- Take average weight of forage of each field
- Multiply with 360…to get apprx.yield /acre
- Approx /expected yield is **15-20 MT/acre**

1 acre=4840 sq yards=43560 sft=(121x360)

forage yield depends on many factors…

1. soil fertility
2. irrigation
3. type of seed
4. season
5. spacing (ie no of plants per acre)

PREPARATION OF PIT

polyethylene sheet is to be laid on the floor, and all the four sides of the silo pit to prevent moisture from the ground and sides.

Wilting ..a day before ensiling

MOVING …CHAFF CUTTER TO THE PIT PLACE

PREPARATION OF ADDITIVE MIXTURE

for each MT of forage add the following in 10 liter of water.. (ie in 10 ml per one kg forage)

@ 1 kg molasses

@ 1 kg salt

@ 1 kg DCP

@ 1 gram LAB

(Kemlac HD inoculant)

HOW TO ADD ADDITIVES MIXTURE

- if 10 plants are kept at a time in feeder of chaff cutter and if each plant is weighing 1 kg….
- total weight will be 10 kgs
- add 100 ml (@10 ml /1 kg forage)of additive mixture to the forage in feeder by pouring with a measuring glass, while moving for chopping

Start filling the pit

CHOPPING SIZE IS IMPORTANT@1 INCH


Keep a person inside to spread chopped forage evenly and for pressing for compaction with legs

PROPERLY COMPACT TO CREATE ANAEROBIC CONDITION

after compacting to create anaerobic condition.....

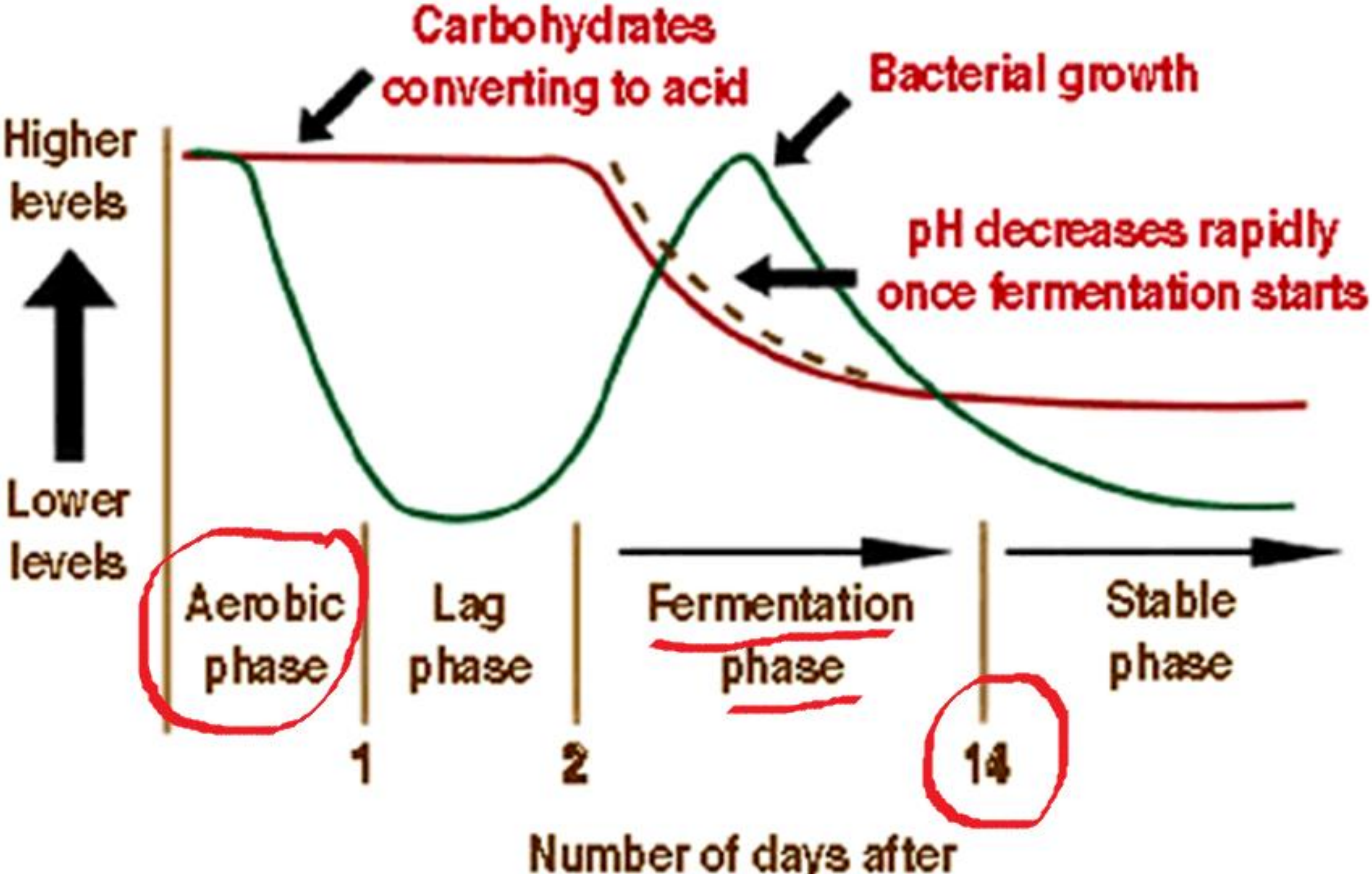
pit is to be sealed immediately, covering with 200 gsm polyethylene sheet

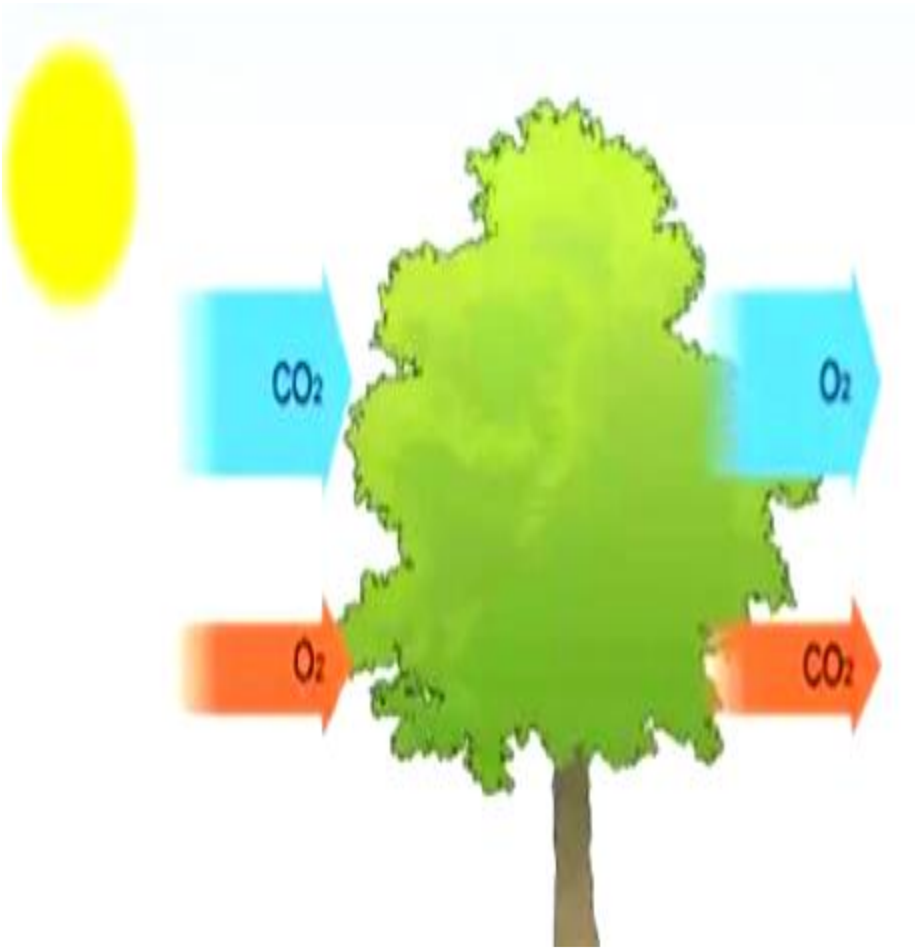
after SEALING SILOPIT

keep heavy weights like granite stones

silo pit is to be protected from rain, sun and wind by keeping asbestos or steel sheets as shown below in the picture.

Fermentation phase





It is not just animals that *respire* – plants carry out respiration as well. Plants respire all the time because their cells need energy to stay alive, but plants can only photosynthesise when they are in the light.

Time of day	Photosynthesis	Respiration
Day	Active	Active
Night	None	Active

FERMENTATION END PRODUCTS

PH	3.7 -4.2
LACTIC ACID	4 - 7
ACETIC ACID	1-3
PROPIONIC ACID	<0.1
BUTYRIC ACID	**0**
ETHANOL	1-3
AMMONIA-N%	5 - 7

NUTRIENT VALEUS OF CORN SILAGE

NUTRIENT	AVEAGE VALUE	RANGE
DRY MATTER	35	35 - 40%
CRUDE PROTEIN	8.0	6-17
ACID DETERGENT FIBER	28.0	20-40
NEUTRAL DETERGENT FIBER	48.0	30-58
TOTAL DIGESTIBLE NUTRIENTS	67.0	55-75
CALCIUM	0.26	0.10-0.40
PHOSPHORUS	0.30	0.10-0.40

All values are percent of dry matter basis.

on the first day of opening of pit ...

pink colored fungus found on top layer & on corners

on the first day of opening of pit …

pink colored fungus found on top layer & on corners

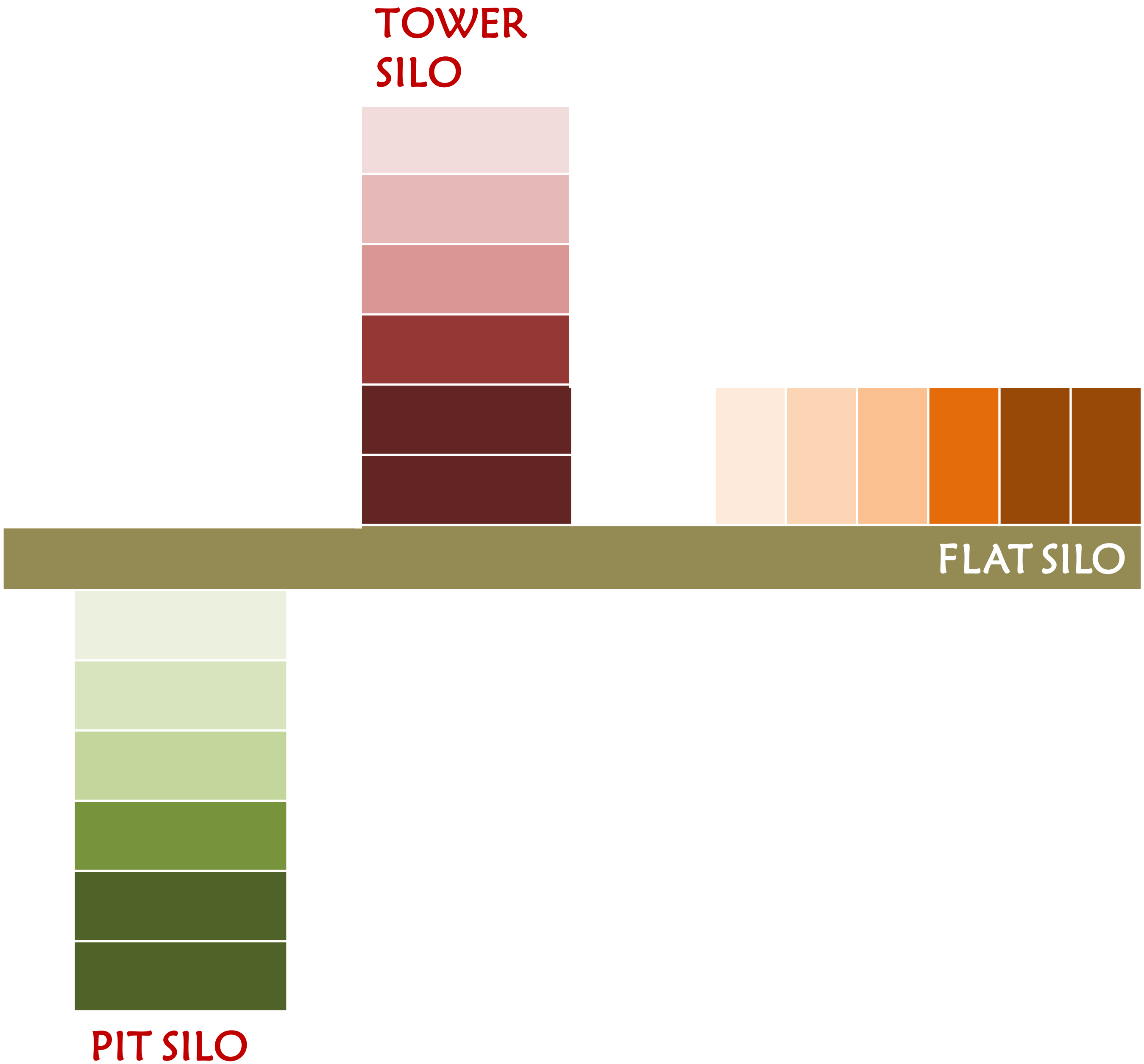
on the first day of opening of pit …

pink colored fungus found on top layer & on corners

Good quality silage looks like .....

golden yellowish and smells ripened fruit like

FEED OUT STAGE


TOWER SILO
FLAT SILO
PIT SILO

taking out day requirement & filling in bags

BROWN SHEEP EATING SILAGE

quality of silage is indicated by

1. VDMI….(voluntary dry matter intake)
2. PRODUCTION
3. FAECAL CONSISTENCY

REASONS FOR REDUCED QUALITY at ensiling

- more moisture in forage
- Slow silo filling
- forage, not chopped finely enough
- inadequate compression & packing
- allowing air to enter stored silage

A PIT, WHERE THE TOP OF PIT IS NOT PROTECTED FROM RAIN

SPOILED SILAGE
DUE TO ENTRY OF RAIN WATER

SPOILED SILAGE
DUE TO LOW QUALITY POLYETHELENE SHEET

REASONS FOR SPOILAGE OF SILAGE at feedout stage

- If the silage is not removed with min 4 inch thickness like a bread slice
- If the pit is not sealed properly after taking out silage
- If the top layer is exposed for more than 30 minutes a day
- If silage grade polyethylene sheet is not used
- If the sheet is not spread completely on all sides

maintenance of silage pit
at feed out stage is
very difficult
than making silage

IF THE SILAGE IS NOT COLLECTED FROM TOP LAYER

CHECKING THE QUALITY OF SILAGE EVERYDAY IS VERY IMPOTANT BEFORE FEEDING TO ANIMALS


Thank You!

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